Deliverable 1: Project Proposal

Cybersecurity Analytics Class Project

Deliverable	*Deliverable 1: Project Proposal*	**Selected Track**	*Fraud*
Project Title	*Deepfake Recognition*	**Team Name**	*Team 6*
Team Members	*Flemming Grabowski, Payel Sarkar*	**Submission Date**	*December 6, 2026*
Repository	*IS665-Team6-Deepfake-Recognition*	**Dataset Link**	*https://www.kaggle.com/datasets/xdxd003/ff-c23*

1. Problem Definition and Security Significance

Deepfakes are fake or edited images and videos created with artificial intelligence. They can make a person appear to say or do something that never happened. Criminals may use deepfakes to impersonate family members or company leaders and trick people into sending money. They may also place someone's face into a harmful image, use it for blackmail, or upload it to an unsafe website. The FBI has warned that AI-generated media is being used for financial fraud, identity impersonation, and extortion (FBI Internet Crime Complaint Center, 2024).

We propose a convolutional neural network (CNN) that will check facial images and short videos and classify them as likely real or fake. Individuals and security teams could use the result as an early warning. It may help someone stop before sending money, sharing personal information, or responding to a threat. The model will support human review. It will not be treated as legal proof or as a guarantee that an image is real.

***Project question:** Can a CNN identify deepfake facial images and videos while keeping missed fakes and false warnings reasonably low?*

2. Project Scope and Success Criteria

The prototype will accept a facial image or short video. For a video, it will select a few frames, find the main face, and check each frame. It will combine the results into one final score. The project will focus only on visual deepfakes. It will not examine cloned voices, identify the person, search websites, remove online content, or determine who created the deepfake.

Success criterion	Target or decision rule	Why it matters
Detect manipulated media	*Deepfake recall of at least 0.8 on the test set that wasn't used during training*	*Missing a fake may expose a user to fraud, impersonation, or blackmail.*
Limit false accusations	*False-positive rate at or below 0.2*	*A real image should not be labeled fake without sufficient evidence.*
Explain the warning	*Show frame scores and a simple face heatmap*	*Users should see what influenced the warning.*

3. Proposed Dataset

We will use FaceForensics++, a research dataset made from 1,000 original videos and four types of face manipulation. We will use a sample of about 200 real and 200 fake videos. We will take about 10 frames from each video, giving us roughly 4,000 face images. We will report the final number after checking the files.

We will divide the videos before taking frames from them. All frames from the same original video will stay in one group. This prevents the model from seeing nearly identical images during both training and testing.

Field	Planned value
Primary dataset	*FaceForensics++*
Unit of analysis	*Training: one cropped facial frame. Inference: one still image or short video*
Target / anomaly	*Binary label: real or fake/manipulated*
Expected class balance	*Approximately 50% real and 50% manipulated in the selected subset.*
Split strategy	*Grouped 70/15/15 train, validation, and test split by source video before frame extraction.*

4. Proposed Technical Approach

We will check the labels and remove missing, unreadable, duplicate, or poor-quality images. Even though the dataset has good ratings on Kaggle, we will still review it carefully. We will then select frames from each video, crop the main face, and resize it to 224 × 224 pixels. We will compare the number of real and fake images and check their quality. During training, we may flip, resize, brighten, or compress some images so the model learns from realistic changes. We will complete these steps using a simple python script, but we will also review the processed images manually for quality assurance.

Other than that we will use Keras, OpenCV, scikit-learn, and Streamlit. MobileNetV2 is already trained to recognize useful image patterns, so adapting it should require less time and computing power than building a large model from the beginning. If time permits, we may also explore Ultralytics HUB as another option for training and testing the model.

Stage	Planned method	Evidence to produce
Data preparation	*Verify labels, remove unreadable and duplicate images, split data by source video before frame extraction to prevent leakage, detect and crop the main face, resize images to 224 × 224 pixels, apply augmentation only to training data.*	*Data-quality summary, class-balance table, number of removed files, sample processed images, and a reproducible preprocessing script.*
Baseline	*Compare against a majority-class classifier and a simple CNN trained using the same grouped train/validation/test split.*	*Baseline accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.*
Candidate model(s)	*Fine-tune MobileNetV2 with ImageNet weights. Compare a frozen-backbone version with a partially fine-tuned version. Aggregate frame-level predictions to produce a video-level prediction.*	*Model configuration, training curves, validation results, model-comparison table, and representative Grad-CAM heatmaps.*
Evaluation	*Use the grouped 70/15/15 split by source video. Tune the decision threshold on the validation set only. Evaluate recall, false-positive rate, precision, F1-score, ROC-AUC, and errors by manipulation type and image quality.*	*Final test-results table, confusion matrix, ROC curve, threshold analysis, false-positive and false-negative examples, and error analysis.*

5. Risks, Ethics, and Responsible AI

A detection score can reduce risk only when it is interpreted carefully. A false negative could allow fraudulent content to appear trustworthy, while a false positive could wrongly damage a person's reputation. The interface will therefore use language such as likely manipulated and needs review or a likelihood scale.

Risk	Likelihood / impact	Planned mitigation or test
New deepfake methods	*High: new tools may create different visual signs.*	*Test one unseen manipulation type when possible and clearly state the model's limits.*
False positives and negatives	*High: either error can cause financial or personal harm.*	*Report both error types, tune the threshold on validation data, and require human review.*
Bias	*Medium to high: results may differ by skin tone, age, lighting, or image quality.*	*Compare available groups and clearly report missing demographic information.*
Privacy and harmful media	*High: facial images and abusive content are sensitive.*	*Use only approved research data, restrict access, avoid collecting victim images, and delete uploads after analysis.*
Edited or blurry files	*Medium: cropping, blur, or compression may hide signs of a deepfake*	*Test resized and compressed images and avoid claiming that the model catches every fake.*
Difficult model decisions	*Medium: CNN results may be hard to explain.*	*Show a confidence score and, if possible, a heatmap of the areas the model focused on.*
Computing and schedule	*Medium: video processing and training may be slow.*	*Use a smaller dataset, transfer learning, and saved face crops.*

6. Work Plan and Team Roles

Task / milestone	Owner(s)	Target date	Expected output
Repository setup	*Flemming and Payel*	*Sep 20, 2026*	*Approved access, README, and task board*
Dataset audit and frame preparation	*Payel*	*Sep 27, 2026*	*Quality notes and prepared face images*
Baseline CNN and interim report 1	*Flemming*	*Oct 4, 2026*	*Notebook and baseline results*
Main model and error review	*Flemming and Payel*	*Oct 10, 2026*	*Model comparison and error examples*
Testing, ethics, and interim report 2	*Payel*	*Nov 4, 2026*	*Compression test and risk review*
Prototype and final report	*Flemming*	*Dec 1, 2026*	*Simple demonstration and report draft*
Final review and presentation	*Flemming and Payel*	*Dec 6, 2026*	*Final PDF, slides, video, and repository*

References

Chollet, F. (2017). Xception: Deep learning with depthwise separable convolutions. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (pp. 1251–1258). https://doi.org/10.1109/CVPR.2017.195

Federal Bureau of Investigation, Internet Crime Complaint Center. (2024, December 3). Criminals use generative artificial intelligence to facilitate financial fraud. https://www.ic3.gov/PSA/2024/PSA241203

Rössler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., & Nießner, M. (2019). FaceForensics++: Learning to detect manipulated facial images. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 1–11). https://doi.org/10.1109/ICCV.2019.00009

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In Proceedings of the IEEE International Conference on Computer Vision (pp. 618–626). https://doi.org/10.1109/ICCV.2017.74

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Team Contribution Statement

Team member	Responsibilities and completed work	Evidence / artifact
Flemming Grabowski	*Researched the CNN model and technical approach and helped write and review the proposal*	*Technical-approach section and project proposal*
Payel Sarkar	*Researched the dataset and data-preparation approach and helped write and review the proposal*	*Dataset section and project proposal*

We stayed in touch regularly and worked together using shared documents We divided the work based on our roles and reviewed the proposal together. If the workload becomes uneven, we will discuss it and adjust the tasks so both of us contribute fairly.

AI-Use Disclosure

Tool	How it was used	What the team verified or changed
ChatGPT	*Helped with topic brainstorming, organizing the proposal, and improving wording and grammar*	*We verified the technical information and citations, revised the wording, and confirmed the project scope*
Claude	*Helped review the completed proposal for clarity, organization, and consistency*	*We reviewed the feedback, selected the useful suggestions, and made the final changes ourselves*

Submission Checklist

Check	Requirement
[x]	Filename follows the required naming convention.
[x]	Proposal is 2–3 pages, excluding references and appendices.
[x]	All bracketed prompts and sample text have been replaced.
[x]	Dataset and repository links open correctly.
[x]	Every team member reviewed the final submission.
[x]	Contribution statement and AI-use disclosure are complete.

Rubric Self-Check

Criterion	Points	Where the evidence appears
Problem definition and significance	2	*Page 1*
Dataset and scope	2	*Page 2*
Technical approach and feasibility	3	*Page 2*
Risk and Responsible AI planning	2	*Page 3*
Work plan, clarity, and completeness	1	*Page 3*